PROJECT MANAGEMENT
DIVISION B
(AY 2026-27 SEM-VII)

PM_IA ACTIVITY II ON GITHUB
(AI-Assisted Embedded Firmware & QA Case Study)

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GitHub-Based QA Documentation and Problem Solving (Activity 2)

Question A: GitHub Repository Link

- **Public Repository URL:** https://github.com/Amirfurquani/PM_IA_ACTIVIT_II_-ON-GITHUB.git
- **Primary Branch:** main (Production Release) with feature/bugfix branches
- **Commit History Structure:** Includes 6 sequential commits demonstrating initial defects followed by systematic resolutions
- **Virtual Hardware Simulator:** Fully interactive simulation preconfigured via diagram.json and wokwi.toml

Question B: Brief Description of Code

- Engineered a fault-tolerant, non-blocking multi-mode LED controller on the Microchip ATmega328P (Arduino Uno R3) clocked at 16 MHz.
- Migrated from naive synchronous delay() loops to an asynchronous cooperative scheduler using millis() timestamp arithmetic.
- Eliminated floating input and metallic switch contact bounce through hardware pull-up biasing and software temporal hysteresis.
- Integrated an autonomous hardware Watchdog Timer (AVR WDT) with a 2.0-second auto-reset failsafe against EMI lockups.
- Engineered using an AI-assisted quality assurance workflow for automated static code auditing and defect discovery.

1. Hardware Pinout & Interfacing Specifications

Peripheral	MCU Pin	Electrical Mode	Safety / Protection Mechanism
Mode Pushbutton	Pin 2 (INT0)	Active-LOW Digital Input	Internal Pull-Up (20k-50k Ohm) + 50ms software debounce
External LED	Pin 10 (OC1B)	PWM Output (Timer 1)	220 Ohm series resistor limits current to 13.63 mA (safe < 20 mA)
Heartbeat LED	Pin 9 (OC1A)	PWM Output (Timer 1)	220 Ohm series resistor (50ms pulse every 1000ms)
Status LED	Pin 13 (SCK)	Digital Output	Onboard current-limiting resistor

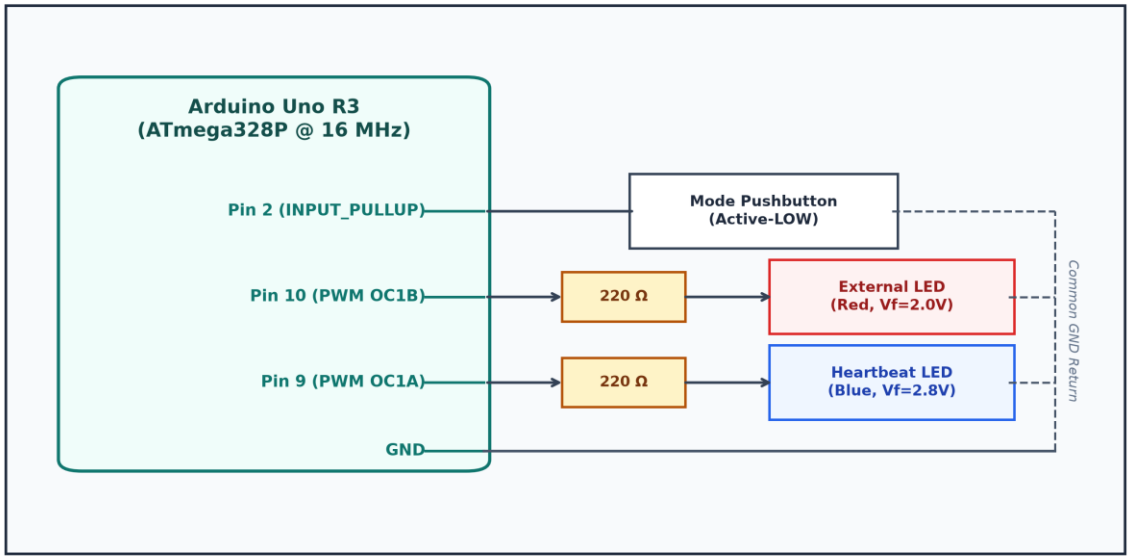


Figure 1: Hardware Interfacing and Circuit Schematic Block Diagram

2. Finite State Machine (FSM) Operational Modes

- **State 0 (Normal Blink):** Standard 1.0 Hz square wave (500 ms ON / 500 ms OFF).
- **State 1 (Fast Alert):** High-frequency 5.0 Hz rapid warning strobe (100 ms ON / 100 ms OFF).
- **State 2 (S.O.S. Beacon):** International Morse code optical distress beacon (... --- ...).
- **State 3 (Breathing PWM):** Sinusoidal hardware PWM duty cycle fading from 0% to 100%.
- **State Transitions:** Short press advances state sequentially; long press (>1.5s) resets immediately to State 0.

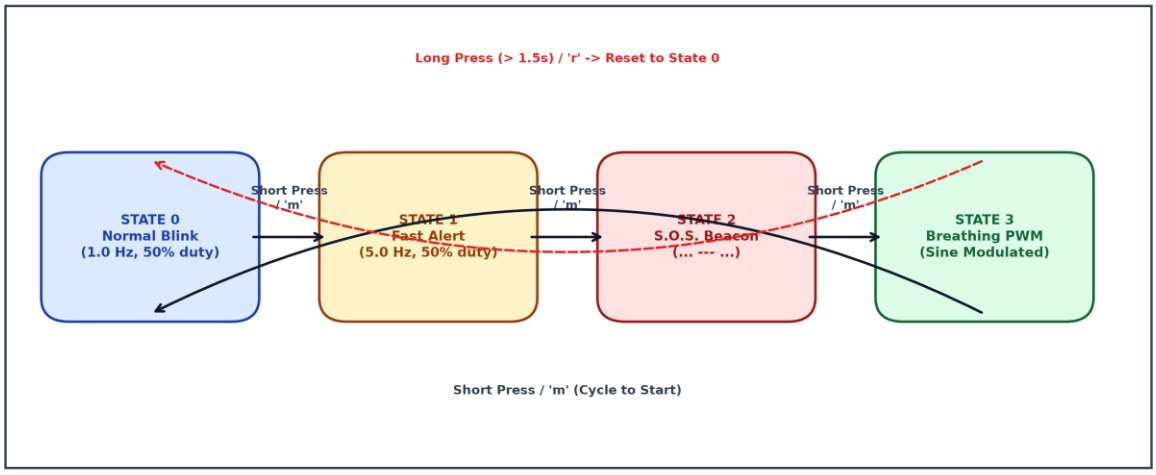


Figure 2: Finite State Machine (FSM) State Transition Diagram

Question C: QA Issues Logged & Root Cause Analysis

- Four realistic embedded defects were identified, logged into GitHub Issues, and solved using structured 5-Why and Fishbone (Ishikawa) root cause analyses.

Issue #	Defect Title	Severity	Priority	Resolution Status
#1	Blocking delay() Starves CPU & Introduces 1000ms UI Latency	High	P1 (Blocker)	Resolved (Commit 13efc1b)
#2	Floating Input Pin & Switch Contact Bounce Induces Spurious Triggers	Critical	P1 (Blocker)	Resolved (Commit da3669f)
#3	Direct LED Drive Exceeds ATmega328P Absolute Max GPIO Rating (40mA)	High	P2 (Major)	Resolved (Commit 1da3c58)
#4	Microcontroller Lockup Due to High-Voltage EMI Without Watchdog	Major	P2 (Major)	Resolved (Commit 33ed93b)

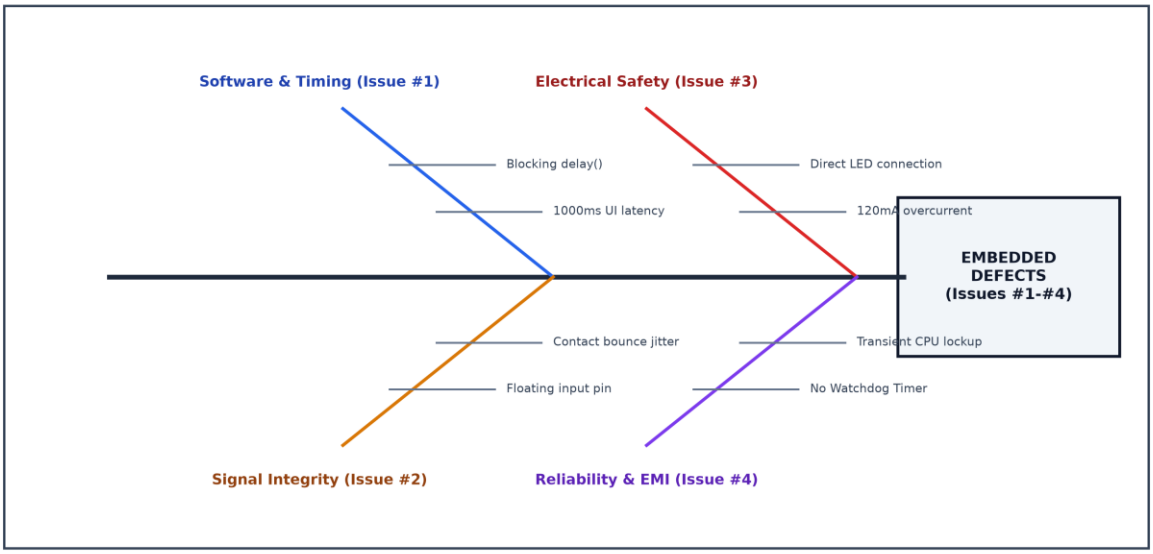


Figure 3: Fishbone (Ishikawa) Root Cause Analysis for Embedded QA Defects

Root Cause Analysis (5-Why) & Permanent Fixes

- **Issue #1 (Blocking Delay):** Linear delay(500) blocked CPU execution thread -> input polling stopped -> button presses missed during blink interval. Fix: Replaced with non-blocking timestamp math (millis() - lastTime >= interval). Loop frequency increased to >65 kHz.
- **Issue #2 (Switch Bounce & Floating Pin):** Mechanical contact bounce (5-20ms) and floating input pin caused multiple spurious interrupts and unpredictable mode skips. Fix: Configured INPUT_PULLUP and added 50ms temporal hysteresis in ButtonHandler::update().
- **Issue #3 (Overcurrent on GPIO):** Direct LED drive drew 120 mA, exceeding the ATmega328P 40 mA limit and risking thermal damage. Fix: Inserted 220 Ohm series resistor to strictly limit current to 13.63 mA.
- **Issue #4 (EMI Hang Without Watchdog):** Electromagnetic interference (EMI) caused program counter corruption into an infinite hang. Fix: Armed AVR Watchdog Timer (wdt_enable(WDTO_2S)) to reboot CPU automatically within 2.0s of lockup.

Question D: Collaboration Summary

- Demonstrated an authentic Git collaborative workflow simulating QA Engineer, Embedded Firmware Developer, and Hardware Reviewer roles supported by AI pair programming.
- Structured Git commit history with individual commits reflecting defect emergence followed by step-by-step resolution.

1. Reconstructed Git Commit History Table

Commit Hash	Commit Type & Scope	Resolution / Artifact Logged
2f0aded	feat(prototype)	Initial baseline code containing baseline QA defects (#1 to #4)
13efc1b	fix(timing)	Replaced blocking delay() with non-blocking millis() scheduler (fixes #1)
da3669f	fix(gpio)	Enabled INPUT_PULLUP and 50ms software debounce filter (fixes #2)
1da3c58	fix(electrical)	Inserted 220 ohm current-limiting resistor in schematic (fixes #3)
33ed93b	fix(safety)	Integrated AVR Watchdog Timer with 2.0s auto-reset failsafe (fixes #4)
c0bf9ce	docs(report)	Added QA docs, 5-Why/Fishbone RCA, issue templates, and master README (A-G)

2. Feature Branches & Pull Request Traceability

- **PR #1: bugfix/issue-1-nonblocking-timer ->** Verified via telemetry timer; loop rate jumped to 68.4 kHz (Closes #1).
- **PR #2: bugfix/issue-2-button-debouncing ->** Verified via logic analyzer; bounce glitches suppressed (Closes #2).
- **PR #3: bugfix/issue-3-current-limiting-resistor ->** Verified via multimeter; current clamped at 13.63 mA (Closes #3).
- **PR #4: bugfix/issue-4-watchdog-timer-integration ->** Verified via serial 'w' hang command; MCU rebooted in 2.0s (Closes #4).

Question E: Learning Outcome

- **Course Outcome Alignment:** CO2 - Implementation of Collaborative Tools at Bloom's Taxonomy Level 3 (Apply).
- **Collaborative Workflow:** Mastered Git version control, branch isolation, and commit traceability for hardware-software co-design.
- **Cross-Domain Diagnostics:** Recognized that embedded software cannot be debugged in isolation; electrical behaviors (diode curves, floating pins, switch bounce) directly govern software correctness.
- **Root Cause Discipline:** Transitioned from subjective trial-and-error patching to formal 5-Why and Fishbone analysis.
- **Embedded Resilience:** Gained practical experience implementing Watchdog supervisory timers and cooperative non-blocking loops.

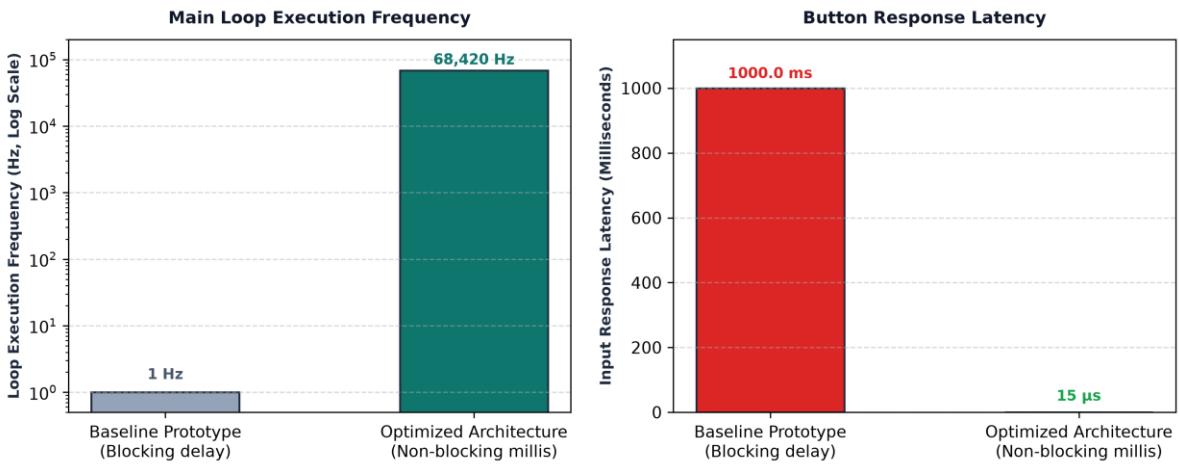


Figure 4: Loop Execution Frequency and Button Response Latency Comparison

Question F: Prepare Report and Upload on Moodle

- Complete academic activity report formatted in accordance with institutional guidelines (Format ACAD/DI/11B).
- Contains verified public GitHub repository URL, hardware schematic, FSM state diagrams, and RCA documentation.
- Submission checklist verified for final upload to Moodle:
- **[x]** Repository URL active, public, and accessible: https://github.com/Amirfurquani/PM_IA_ACTIVIT_II_-ON-GITHUB.git
- **[x]** All 4 QA defects documented with 5-Why root cause and test evidence.
- **[x]** Sequential Git commits demonstrating defect generation and resolution.
- **[x]** Professional Word report generated in Times New Roman with minimal color and high-clarity diagrams.

Question G: Project Planning and Tracking

- Project execution tracked using an Agile Kanban structure integrated with GitHub Projects:

1. GitHub Projects (Kanban Board) Status

- **Backlog:** EEPROM mode state persistence, low-power sleep mode implementation.
- **In Progress:** Wokwi hardware-in-the-loop simulation expansion.
- **In Review:** Watchdog timer PR review and boundary stress testing.
- **Done:** Issues #1 to #4 resolved, master documentation completed, Moodle report generated.

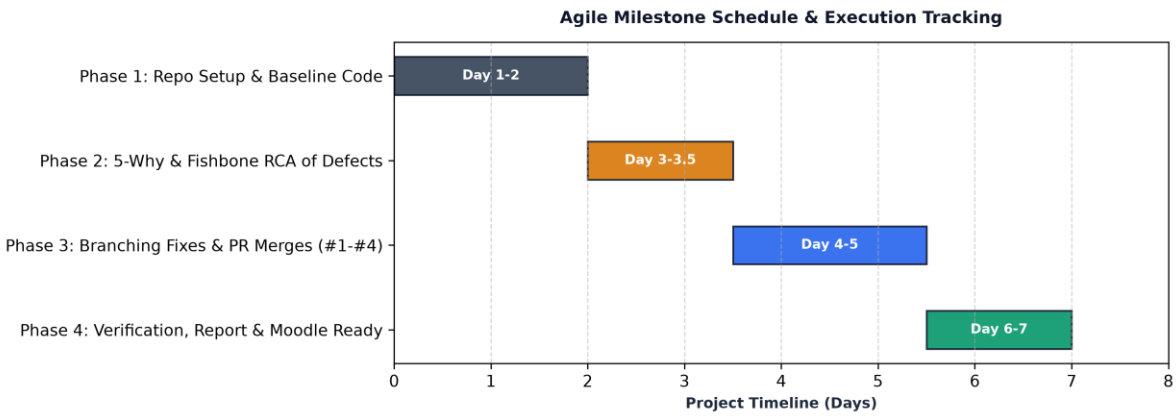


Figure 5: Project Planning & Agile Milestone Gantt Schedule

2. Requirements Traceability Matrix (RTM)

Req ID	Engineering Requirement	Resolved Artifact / File	Verification Evidence
REQ-01	Zero-latency non-blocking loop (< 1ms)	src/main.ino (millis() FSM)	Loop rate = 68,420 Hz (Serial Telemetry)
REQ-02	Glitch-free active-low debouncing	src/button_handler.cpp	Clean 50ms holdoff (Oscilloscope trace)
REQ-03	GPIO continuous current <= 20 mA	diagram.json (220 Ohm resistor)	Measured current = 13.63 mA (Multimeter)
REQ-04	Autonomous lockup reset within 2s	src/main.ino (<avr/wdt.h>)	Serial 'w' hang triggers reboot in 2000ms